# Lijie Yao 姚李捷

**Ph.D. M.Sc. Ing. B.Eng.**

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**Languages Spoken:** Chinese (native fluency) | English (advanced proficiency) | French (advanced proficiency)

## ABOUT ME

I am an **Assistant Professor** and **Programme Director of Computer Science and Technology** at **Xi'an Jiaotong-Liverpool University (XJTLU)**. I obtained my Ph.D. degree from **Université Paris-Saclay (UPS)**. My research interests include *visualization in motion*, *AI-assisted data communication*, and *human-AI interaction*. My work received the **Ph.D. Thesis Prize** from **ACM AFIHM** and **Canada Mitacs Globalink Research Award**. Part of my work on *visualization in motion* was applied in the **2024 Paris Olympics** swimming races. Previously, I was a postdoctoral and doctoral researcher at **INRIA** and a visiting researcher at **University of Calgary**. Collaboration emails are more than welcome.

## EDUCATION

**Université Paris-Saclay** Gif-sur-Yvette, France Sept. 2020 – Dec. 2023
Doctor of Philosophy – Computer Science

- Supervisors: Petra Isenberg & Anastasia Bezerianos
- Thesis Topic: Situated Visualization in Motion
- Research Focus: Visualization in motion, perception, visual design

**Université Paris-Saclay** Gif-sur-Yvette, France Sept. 2019 – Aug. 2020
Master of Science – Computer Science

- Supervisors: Petra Isenberg
- Thesis Topic: Situated Visualization in Motion
- Research Focus: Visualization in motion, perception, visual design

**Polytech Lille** Lille, France Sept. 2017 – Aug. 2019
Diplôme d'ingénieur (= Master of Engineering) – Electrical Engineering and Automation

- Supervisors: Komi Midzodzi Pekpe
- Thesis Topic: Position Control of A Drone
- Research Focus: Embedded systems, automatic control

**XIDIAN University | 西安电子科技大学** Xi'an, China Aug. 2014 – June 2018
Bachelor of Engineering – Electronic Information Engineering

- Supervisors: Xavier Redon & Alexandre Boé
- Thesis Topic: Electronic Necklace for Health Monitoring
- Research Focus: Embedded systems, microelectronics

## WORK EXPERIENCE

**Assistant Professor** Xi'an Jiaotong-Liverpool University July 2024 – Now
Suzhou, Jiangsu, China

- Xi'an Jiaotong-Liverpool University is a Chinese private university founded in 2006 as a joint venture between the University of Liverpool and Xi'an Jiaotong University.
- My duties include research, teaching, and academic/administrative services.

**Postdoctoral Researcher** Inria Jan. 2024 – March 2024

Gif-Sur-Yevette, Île-de-France, France

- Inria is a French national institution for research in digital science and technology. It is ranked 18th in the global research institution and 7th in computer science.
- My duties included reviewing literature, conducting experiments, and writing papers.

**Visiting Researcher** University of Calgary July 2023 – Oct. 2023

Calgary, Alberta, Canada

- The University of Calgary is a Canadian public research university founded in 1908, prior to being instituted into a separate, autonomous university in 1966.
- My duties included reviewing literature and conducting experiments.

**Teaching Assistant** Université Paris-Saclay Sept. 2021 – Aug. 2023

Gif-Sur-Yevette, Île-de-France, France

- The Université Paris-Saclay is a french combined technological research institute and public research university. It is ranked 13$^{th}$ in the ARWU 2025 and 70$^{th}$ in QS 2026.
- My duties included reviewing literature and conducting experiments.

**Doctoral Researcher** Inria Sept. 2020 – Dec. 2023

Gif-Sur-Yevette, Île-de-France, France

- Inria is a French national institution for research in digital science and technology. It is ranked 18$^{th}$ in the global research institution and 7$^{th}$ in computer science.
- My duties included reviewing literature, conducting experiments, writing papers, advising students, and presenting.

## PEER-REVIEWED PUBLICATIONS

* Corresponding author, † These authors contribute equally. List order: reverse chronology.

### Journals

1. Junhao Zhao, Romain Vuillemot, Petra Isenberg, **Lijie Yao***. Authoring Narrative Visualization in Motion: Visual Storytelling in Swimming Videos. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2027. In press.
2. Fairouz Grioui*, **Lijie Yao**, Petra Isenberg, Tanja Blascheck. A Smartwatch Moves: The Impact of Walking Speed and Trajectories on Micro Visualizations Reading on a Smartwatch. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2027. DOI: 10.1109/TVCG.2026.3735571. In press.
3. Charles Perin*, Charles-Olivier Dufresne-Camaro, Maxime Cordeil, Jo Jo Gong, Christophe Hurter, Purang P. Irani, Jade Kandel, Ang Li, Tica Lin, Ben Pearman, Yalong Yang, **Lijie Yao**, Wesley Willett. Opportunities for First-Person Visualization for Sports and Physical Activities. *IEEE Computer Graphics and Applications (CG&A)*, 2027. In press.
4. Junxiu Tang, **Lijie Yao***, Lu Ying, Romain Vuillemot, Petra Isenberg. Diving Deep into Time: Temporal Arrangements for Embedded Visualization in Swimming Videos. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2026, 32(7), pp.6825-6841. DOI: 10.1109/TVCG.2026.3689361.
5. **Lijie Yao***, Federica Bucchieri, Victoria McArthur, Anastasia Bezerianos, Petra Isenberg. User Experience of Visualizations in Motion: A Case Study and Design Considerations. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025, 31 (1), pp.174-184. DOI: 1109/TVCG.2024.3456319.
6. **Lijie Yao***. Situated Visualization in Motion. *IEEE Computer Graphics and Applications (CG&A)*. 2024, 44(6), pp.142-150.

DOI: 10.1109/MCG.2024.3462129.

7. **Lijie Yao***, Romain Vuillemot, Anastasia Bezerianos, Petra Isenberg. Designing for Visualization in Motion: Embedding Visualizations in Swimming Videos. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2024, 30 (3), pp.1821-1836. DOI: 10.1109/TVCG.2023.3341990.
8. **Lijie Yao***, Anastasia Bezerianos, Romain Vuillemot, Petra Isenberg. Visualization in Motion: A Research Agenda and Two Evaluations. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2022, 28(10), pp.3546-3562, DOI: 10.1109/TVCG.2022.3184993.

**Conferences**

9. Jinyu Cai†, Xinling Du†, Guang Yang, Liu Yang, **Lijie Yao***. EduAgent: A Visual Feedback Agent for Teaching Python to Novices. *IEEE VIS workshop on GenAI, Agents, and the Future of VIS*, Nov. 2026, Boston, U.S.
10. Xi Zhou, Junhao Zhao, Liu Yang, **Lijie Yao***. Enriching Practice Understanding for Leisure-Time Squash Players. *Posters of the IEEE VIS*, Nov. 2026, Boston, U.S.
11. Xichun Xu, Ruofei Du, Jiaqi Zheng, Qingping Wang, **Lijie Yao**, Liu Yang*. Beyond Turns: Fragment-Response Alignment for Evolving Intent and Traceable Review in LLM Chat. *Posters of the IEEE VIS*, Nov. 2026, Boston, U.S.
12. **Best Paper Honorable Mention** Jingyu Wang, Jiahan Zhang, Yu Liu, Lingyun Yu, **Lijie Yao***. Tangible Carbon Blocks: Data Physicalization for Everyday Carbon Emissions. *ChinaVis*, July 2026, Guiyang, China.
13. Zehong Ye, Jingyu Wang, Xiao Xie, Lingyun Yu, **Lijie Yao***. When Color Contrast Is Not Enough: An Empirical Study of Dynamic Color Adaptation for In-game Health Visualizations. *ChinaVis*, July 2026, Guiyang, China.
14. Wanfang Xu, Jifan Yang, Fengwen Zhang, Yu Lu, **Lijie Yao**, Le Liu, Lingyun Yu*. Touch, Sound, and Space: Exploring Immersive Music Interaction through AI-Generated Environments. *International Symposium on Visual Information Communication and Interaction (VINCI)*, Dec. 2025, Linz, Austria, Article 23, pp. 1-8. DOI: 10.1145/3769534.3769589.
15. Haonan Yao, Lingyun Yu, **Lijie Yao***. Designing Visualization Widgets for Tangible Data Exploration: A Systematic Review. *IEEE Visualization & Visual Analytics (IEEE VIS)*, Nov. 2025, Vienna, Austria, pp. 261-265. DOI: 10.1109/VIS60296.2025.00058.
16. **Lijie Yao***, Yuming Zhang, Yu Liu. Input Physicalization in Practice: An Instructional Walkthrough with Visualization Novices. *IEEE VIS workshop on Input Visualization*, Nov. 2025, Vienna, Austria. Link: osf.io/dfgvx
17. Qi Guo†, Junyi Li†, Jiayi Hong, **Lijie Yao***. Early Exploration into AI-Assisted Visual Analytics for Dynamic Videos. *IEEE VIS workshop on GenAI, Agents, and the Future of VIS*, Nov. 2025, Vienna, Austria. Link: visxgenai.github.io/subs-2025/1558/1558-doc.pdf
18. Junhao Zhao, **Lijie Yao***. Structured AI Agents for Reliable Visualization Report Generation. *IEEE VIS workshop on GenAI, Agents, and the Future of VIS*, Nov. 2025, Vienna, Austria. Link: nemozjh.github.io/VIS_Agent_nemo/
19. **Best Poster Honorable Mention** **Lijie Yao***. Visualization in Motion: Perception, Design, and User Experience. *Posters of ChinaVis*, July 2025, Hangzhou, China.
20. Fairouz Grioui*, Tanja Blascheck, **Lijie Yao**, Petra Isenberg. Micro Visualizations on a Smartwatch: Assessing Reading Performance While Walking. *IEEE Visualization & Visual Analytics (IEEE VIS)*, Oct. 2024, Florida, U.S. DOI: 10.1109/VIS55277.2024.00017.
21. Ahmed Elshabasi*, **Lijie Yao**, Petra Isenberg, Charles Perin, Wesley Willett. Collecting Information Needs for Egocentric Visualizations while Running. *IEEE VIS workshop on First-Person Visualizations for Outdoor Physical Activities*, Oct. 2024, Florida, U.S. DOI: 10.48550/arXiv.2409.06426.
22. Yvonne Jansen, Federica Bucchieri, Pierre Dragicevic, Martin Hachet, Morgane Koval, Léana Petiot, Arnaud Prouzeau, Dieter Schmalstieg, **Lijie Yao**, Petra Isenberg*. Envisioning Situated Visualizations of Environmental Footprints in an Urban Environment. *IEEE VIS workshop on Visualization for Social Good*, Oct. 2022, Oklahoma City, U.S. DOI: 10.5281/zenodo.7053934.

23. Alaul Islam†, **Lijie Yao***†, Anastasia Bezerianos, Tanja Blascheck, Tingying He, et al.. Reflections on Visualization in Motion for Fitness Trackers. *ACM MobileHCI workshop on New Trends in HCI and Sports*, Sep. 2022, Vancouver, Canada. DOI: 10.48550/arXiv.2409.06401.
24. Federica Bucchieri*, **Lijie Yao**, Petra Isenberg. Situated Visualization in Motion for Video Games. *Posters of the EuroVis*, Jun 2022, Rome, Italy. DOI: 10.2312/evp.20221119.
25. **Lijie Yao***, Anastasia Bezerianos, Romain Vuillemot, Petra Isenberg. Situated Visualization in Motion for Swimming. *Posters of Journée Visu 2022*, Jun 2022, Bordeaux, France. DOI: 10.48550/arXiv.2409.07695.
26. Federica Bucchieri*, **Lijie Yao**, Petra Isenberg. Visualization in Motion in Video Games for Different Types of Data. *Posters of JournéeVisu 2022*, Jun 2022, Bordeaux, France. DOI: 10.48550/arXiv.2409.07696.
27. **Lijie Yao***, Anastasia Bezerianos, Petra Isenberg. Situated Visualization in Motion. *Posters of the IEEE VIS*, Oct 2020, Salt Lake City, U.S.. 2020. DOI: 10.48550/arXiv.2409.07005.

**Selected Proposals**

28. Ang Li*, **Lijie Yao**, Yi Li, Linjia He, Tica Lin, Yalong Yang, Charles Perin, Maxime Cordeil. First-Person eXeRtion (FPXR): XR Visualization and Interaction in Motion. *Selected proposals of workshop at IEEE VR*, Feb. 2027, Melbourne, Australia.
29. Charles Perin*, Tica Lin, **Lijie Yao**, Yalong Yang, Maxime Cordeil, Wesley Willett. First-Person Visualizations for Outdoor Physical Activities: Challenges and Opportunities. *Selected proposals of workshop at IEEE VIS*, Oct. 2024, Florida, United States.
30. **Lijie Yao***. Situated Visualization in Motion. *Selected proposals of Doctoral Colloquium at IEEE VIS*, Oct. 2022, Oklahoma City, United States.

**Thesis**

31. **Lijie Yao***. Situated Visualization in Motion. Human-Computer Interaction [cs.HC]. Université Paris-Saclay, 2023. English. NNT: 2023UPASG093. URL: theses.hal.science/tel-04413122.

## RESEARCH ACTIVITIES

**Grants**

- Young Scientist program of the National Natural Science Foundation of China (NSFC) | 国家自然科学基金青 C (PI, Grant № 62602522) 01/01/2027–12/31/2029
- Medical-Engineering Interdisciplinarity Research Grants, Ruijin Hospital | 上海交通大学医学院附属瑞金医院“青年培育计划医工交叉”项目 (Co-PI, Grant № 2026JC080) 07/2026–07/2027
- Open Project Program of the State Key Laboratory of CAD&CG, Zhejiang University | 浙江大学 CAD&CG 全国重点实验室开放课题 (PI, Grant № A2514) 01/01/2025–31/12/2026
- XJTLU RDF (PI, Grant № RDF-24-01-062) | 西交利物浦大学研究发展基金 01/01/2025–31/12/2027

**Awards**

- Best Paper Honorable Mention at ChinaVis July 2026
- Jiangsu Provincial Double-Innovation Doctor Program | 江苏省“双创博士” (№ JSSCBS2025067) Dec. 2025
- Best Poster Honorable Mention at ChinaVis July 2025
- ACM AFIHM Ph.D. Thesis Prize March 2024
- Canada Mitacs Globalink Research Award May. 2023

**Organization**

- 2028:
  Organizing Committee (Web Chair) , *International Eurographics Conference on Visualization (EuroVis)*, TBD, Paris, France.

- 2027:
  Organizing Committee (Community/Meetups/Election Chair) , *IEEE Visualization & Visual Analytics (IEEE VIS)*, Nov. 9-13, Chichago, United States.
  Co-organizer for workshop on First-Person eXeRtion (FPXR): XR Visualization and Interaction in Motion, *IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, Feb. 27, Melbourne, Australia.
- 2026:
  Organizing Committee (Community/Meetups/Election Chair) & Program Committee (Full papers & Posters), *IEEE Visualization & Visual Analytics (IEEE VIS)*, Nov. 9-13, Boston, United States.
  Program Committee (Technical papers & Art tracks), *International Symposium on Visual Information Communication and Interaction (VINCI)*, Nov. 1-4, Hangzhou, China.
  Program Committee (Full papers & Competition), *ChinaVis*, July 19-22, Guiyang, China.
- 2025:
  Program Committee (Technical papers), *International Symposium on Visual Information Communication and Interaction (VINCI)*, Dec. 1-3, Linz, Austria.
  Program Committee (Full papers & Posters), *IEEE Visualization & Visual Analytics (IEEE VIS)*, Nov. 2-7, Vienna, Austria.
  Program Committee (Short papers), *ACM Symposium on Eye Tracking Research & Applications (ETRA)*, May 26-29, Tokyo, Japan.
- 2024:
  Co-organizer for workshop on First-Person Visualizations for Outdoor Physical Activities, *IEEE visualization & Visual Analytics (IEEE VIS)*, Oct. 13, Florida, United States.

**Review**

- 2027: CHI
- 2026: CHI **Special Recognitions for Outstanding Reviews**, VIS, EuroVis, TVCG, PacificVis (TVCG journal track), ISMAR, ETRA, HRI, JOV, ChinaVis, VINCI, CG&A, JGSA
- 2025: CHI, VIS, EuroVis, TVCG, ISMAR, UIST, ETRA, HRI, NSERC, ChinaVis, IWC, VINCI
- 2024: CHI **Special Recognitions for Outstanding Reviews**, VIS, EuroVis, TVCG, TSMC, ETRA, alt.chi, IMX, HRI, SAIS, ISS, MobileHCI, ETVIS
- 2023: CHI, VIS, EuroVis, CHI LBW
- 2022: VIS, SIGGRAPH Asia

## TEACHING ACTIVITIES

**Grants**

- XJTLU TDF (PI, Grant № TDF2425-R30-283) 01/09/2025–31/08/2027
- XJTLU EASE (Team member, № TDFEASE-R1-2526-42) 01/09/2026–31/12/2026

**Awards**

- Excellent SURF Poster Award × 1, Student-nominated Winner × 1 Oct. 2026
- Third Prize in XIPU “AI in Education” Innovation Competition May 2026
- First-round Certificate in XJTLU Global Dream-Chasers Entrepreneurial Competition April 2026
- Excellent SURF Poster Awards × 2, Student-nominated Winner × 1 Sept. 2025
- Second Prize in XIPU “AI in Education” Innovation Competition Feb. 2025

## Courses Delivery

| Year | Course Name | Level | Student Numbers | Teaching Language |
|---|---|---|---|---|
| | | **Xi'an Jiaotong-Liverpool University** | | |
| 2026–now | Human-AI Interaction (Module leader) | Graduate | ∼30 | English |
| 2024–now | FMP (MRes & MSc) | Graduate | See **Supervision** for details | English |
| 2024–now | Microprocessor system (Module leader) | Undergraduate | ∼600 | English |
| 2024–now | FYP (BSc & BEng) | Undergraduate | See **Supervision** for details | English |
| 2024–2026 | Foundations of Computer Science and Engineering | Undergraduate | ∼720 | English |
| | | **Université Paris-Saclay** | | |
| 2021–2023 | Interactive Information Visualization | Graduate | ∼50 | English |

## SUPERVISION

**Post-graduate**

| Year | Degree-granting Institution | Program | Student (F/M) | Outcomes |
|---|---|---|---|---|
| 2026–now | University of Liverpool | PhD | Junhao Zhao | 1 × TVCG full paper<br>1 × IEEE VIS poster |

**Graduate**

| Year | Degree-granting Institution | Program | Students (F/M, Next Step) |
|---|---|---|---|
| 2026–2027 | University of Liverpool | MSc | Xiaomeng Zhu (F), Zhuoyuan Liu, Huan Wang |
| 2026–2028 | University of Liverpool | MRes | Yuting Li (F) |
| 2025–2026 | University of Liverpool | MSc | Yiming Zhang (Entrepreneurship), Shiyuan Jiang (Public-sector), Zehong Ye (NU-VERSE Games), Yihan Dong (F, BOCHU Electronic), Fanhua Liu (AMD) |
| 2024–2026 | University of Liverpool | MRes | Yuzhu Chen (F, Bank of Communications) |
| 2021–2022 | Université Paris-Saclay | MSc | Federica Bucchieri (F, 3DClouds) |

*Selected Outcomes:* Zehong Ye (1 × ChinaVis full paper); Federica Bucchieri (1 × TVCG full paper, 1 × EuroVis poster, 1 × JournéeVisu poster).

**Undergraduate**

| Year | Degree-granting Institution | Program | Students (F/M, Next Step) |
|---|---|---|---|
| 2026–2027 | XJTLU & University of Liverpool | BSc | Xinling Du (F), Jinyu Cai, Xilai Bu (F), Xun Gong (F) |
| 2026–2027 | XJTLU & University of Liverpool | BEng | Qi Guo (F), Zining Shan (F), Zihan Zhao (F), Yihong Chen |
| 2025–2026 | XJTLU & University of Liverpool | BSc | Ruoxi Liao (F, Master at NUS), Tiantian Ye (F, Master at UCL), Xi Zhou (F, Master at Edinburgh) |
| 2025–2026 | XJTLU & University of Liverpool | BEng | Haochen Yan (Master at UCL), Ziyuan Wang (Master at Edinburgh), Weixin Xiong (F, Master at NUS), Yating Wu (F, Master at UCL), Xiaoran Zhao (F, Master at Manchester), Xiaoyu Li (F, Master at Bristol) |
| 2024–2025 | XJTLU & University of Liverpool | BEng | Xing Wei (Master at Sydney), Yuming Zhang (State-owned enterprise), Yongxuan Teng (Meituan), Dingkang Yang (Master at Monash) |

*Selected Outcomes:* Xi Zhou (1 × IEEE VIS poster); Yuming Zhang (1 × IEEE VIS workshop paper).

**Summer Undergraduate Research Fellowship (SURF)**

| Year | Students (F/M) | Outcomes |
|---|---|---|
| 2026 | Xun Gong (F), Xilai Bu (F), Xibei Chen (F), Yinxuan Liu (F) | Excellent SURF Award |
| | Jinyu Cai, Xinling Du (F) | Student-nominated Winner |
| | Mingcheng Fan, Yujia Kan (F), Ruizhu Liu (F) | |
| 2025 | Qi Guo (F), Junyi Li (F), Ruoxi Liao (F), Jingyu Wang (F), Haochen Yan | Excellent SURF Award; Student-nominated Winner |
| | Xun Gong (F), Ziyi Jin (F), Ziyuan Wang, Weixin Xiong (F) | Excellent SURF Award |

*Selected Outcomes:* Jingyu Wang (2 × ChinaVis full paper); Qi Guo & Junyi Li (1 × IEEE VIS workshop paper).

**Research Assistants**

| Year | Students (F/M) | Outcomes | Next Step |
|---|---|---|---|
| 2026–now | Jingyu Wang (F) | 2 × ChinaVis full paper | |
| 2025–2026 | Junhao Zhao | 1 × IEEE VIS workshop paper | PhD student at Xi'an Jiaotong-Liverpool University |
| 2024–2025 | Haonan Yao | 1 × IEEE VIS short paper | PhD student at University of Nottingham Ningbo China |